SHASHWAT SINGH

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I am currently a PhD student at the University of Glasgow and working on probing massive and supermassive black holes using gravitational wave sources to be detected by LISA – a future space-based gravitational-wave detector and electromagnetic transients from ZTF & LSST.

Academic Qualifications

2023 – 2027 | **University of Glasgow, School of Physics and Astronomy, PhD**
Thesis title : Revealing the family of massive black holes with LISA
Supervisors : Dr Christopher Berry and Dr John Veitch
Gravitational Waves | LISA | Super/Massive Black Holes | Population Inference

2022 – 2023 | **l'Observatoire de Paris, Université PSL (Paris Sciences & Lettres), M2 – International Research Track**
Program combined with courses and research in laboratories.
General-relativity | Data-analysis | Magneto-hydrodynamics | High-performance-computing

2021 – 2022 | **Sciences et Ingénierie, Sorbonne Université, M1 – Paris Physics Masters**
Program targeted towards fundamental courses and compulsory lab-work.
Advanced quantum mechanics | Statistical mechanics | Astrophysics & Cosmology | Numerical methods

2017 – 2021 | **Sardar Vallabhbhai National Institute of Technology, B. Tech, Mechanical Engineering**
Four-year program combined with theoretical and experimental work.
Data-analysis | Fluids - mechanics & dynamics | Machines and rigid body motion

Research Experience

January 2023 – July 2023 | **Max-Planck-Institut für Gravitationsphysik, (Albert-Einstein-Institut), Supervisor : Dr M. Zumalacárregui**
Master Thesis : Probing Fuzzy dark matter using lensed gravitational waves detected by LISA.
Master Thesis | Gravitational-Wave-Lensing | Fuzzy Dark-Matter

September 2022 – January 2023 | **l'Observatoire de Paris, Lab Insertion Unit, Supervisor : Dr A. Hees (SYRTE) & Dr N. Korsakova (APC)**
Waveform compression of Extreme-Mass-Ratios-Inspirals (EMRIs) using Singular Valued Decomposition.
Gravitational-Waves | EMRI | Waveform-modeling

April 2022 – July 2022 | **Max-Planck-Institut für Gravitationsphysik, (Albert-Einstein-Institut), Supervisor : Dr A. H. Nitz**
Prospects of premerger detections & skylocalization of gravitational waves (GWs), using higher modes.
Internship | Gravitational-Waves | Higher-modes | Premerger-detection | Multi-Messenger Astronomy (MMA) | PyCBC

June 2020 – September 2020 | **Max-Planck-Institut für Gravitationsphysik, (Albert-Einstein-Institut), Supervisor : Dr A. H. Nitz**
Build a prototype analysis for massive binary blackhole (MBH) mergers using the LISA.
Internship | Gravitational-Waves | LISA | PyCBC | Simulation

May 2019 – July 2019 | **Dept. of Mechanical Engineering, Indian Institute of Technology Indore, Supervisor : Dr S. K. Sahu**
- Developed numerical method to study heat transfer effects of synthetic jet on different materials in the shape of a 2D plate.
- Developed C++ code for allowing mesh motion within the model in Ansys Fluent.

Internship | Synthetic-jet | Ansys Fluent | Computational-Fluid-Dynamics (CFD) | C++

May 2019 – August 2020

Dept. of Physics, SVNIT Surat, Under supervision of Dr K. N. Pathak

Worked on several projects, especially targeted towards the use of deep learning

- estimating parameters of GWs (Convolutional neural networks)
- sequence-prediction of galaxy mergers (Long-short term memory neural networks)

Gravitational-Waves | Galaxy mergers | Sequence-prediction | Machine Learning (ML)

Publications and Preprints†

- "**Revealing massive black hole astrophysics : The potential of hierarchical inference with extreme mass-ratio inspiral observations**", **Singh, S.**, Chapman-Bird, C E A., Berry, C P L., Veitch, J., https://doi.org/10.3847/1538-4357/ae75dc.
- "**The Early Career Workshop of GR-Amaldi 2025**", *et al.*, https://doi.org/10.1088/1742-6596/3177/1/012001
- "**Constraints on the extreme mass-ratio inspiral population from LISA data**", **Singh, S.**, Chapman-Bird, C E A., Berry, C P L., Veitch, J., https://doi.org/10.1088/1742-6596/3177/1/012113.
- "**Gravitational wave lensing : probing Fuzzy Dark Matter with LISA**", **Singh, S.**, Brando, G., Savastano, S., Zumalacárregui, M., Journal of Cosmology and Astroparticle Physics, Volume 2025, July 2025, https://doi.org/10.1088/1475-7516/2025/07/025.
- "**Estimating dynamical parameters of two interacting galaxies using Deep Learning**", Mahor, A., Reddy, J., Singh, A., **Singh, S.**, Monthly Notices of the Royal Astronomical Society, Volume 521, Issue 3, May 2023, Pages 3441–3450, https://doi.org/10.1093/mnras/stad700
- "**Deep learning for estimating parameters of gravitational waves**", **Singh, S.**, Singh, A., Prajapati, A., Pathak, K. N., Monthly Notices of the Royal Astronomical Society, Volume 508, Issue 1, November 2021, Pages 1358–1370, https://doi.org/10.1093/mnras/stab2417
- "**Lindblad Evolution and Quantum to Classical Transition of Rabi Oscillation in Single Quantum Dot**" Prajapati, A., **Singh, S.** AIP Conference Proceedings 2220, 020122 (2020) ; https://doi.org/10.1063/5.0001258
- "**Experimental and Numerical Investigation of Thermal Performance of Synthetic Jet Impingement**" Singh, P. K., Kothar, R., Sahu, S., Upadhyay, P.K., **Singh, S.**, ICONE2020-16775, V001T03A020; 6 pages, https://doi.org/10.1115/ICONE2020-16775
- "**Experimental and numerical investigation of the thermal performance of impinging synthetic jets with different waveforms**" Singh, P. K., Sahu, S., Upadhyay, P.K., **Singh, S.**, Experimental Heat Transfer, 10.1080/08916152.2021.1984341
- P. K. Singh, A. Kumar, A. Shah, A. Kishor, S. K. Sahu, P. K. Upadhyay, **S.Singh**,"**Flow and Heat Transfer analysis of an axisymmetric Impinging Synthetic Jet for Electronic Cooling**" Proce of Int Conf on Innovation and Thermo-Fluid Eng and Sci [ICITFES – 2020] NIT Rourkela, India, 10-12 February [Paper ID : 13754]
- "**Decoherence Control via Pumping of Electromagnetic Energy in Open Quantum System**" Prajapati, A., **Singh, S**. presented at The 5th International Conference on Atomic, Molecular, Nano-physics with Application (CAMNP-2019).
- P. K. Singh, A. Kumar, A. Shah, A. Kishor, S. K. Sahu, P. K. Upadhyay, **S.Singh**,"**Flow and Heat Transfer analysis of an axisymmetric Impinging Synthetic Jet for Electronic Cooling**" Proce of Int Conf on Innovation and Thermo-Fluid Eng and Sci [ICITFES – 2020] NIT Rourkela, India, 10-12 February [Paper ID : 13754]
- "**Predicting future astronomical events using deep learning**" **Singh, S.**, Prajapati, A., Pathak, K.N. - https://arxiv.org/abs/2012.15476.†
- "**Prospects of detection of lensed gravitational wave signals.**" **Singh S.** (2022). Zenodo. https://doi.org/10.5281/zenodo.7029226.†

Conference & Seminar organization‡ and contributions

- National Astronomy Meeting 2026, University of Birmingham, July 2026. *Probing massive black hole population with combined EMRI and TDE observations*.
- European Astronomical Society Annual Meeting, SwissTech Convention Centre Lausanne, Switzerland, July 2026. *Probing massive black hole population with combined EMRI and TDE observations*.
- Massive Black Hole Spin Workshop, University of Edinburgh, UK, April 2026. *Disrupted stars and falling compact objects : Combining multi-messenger observations to constrain the population of MBH*.
- Student Seminar, University of Glasgow, UK‡, 2024–2025
- GW:UK@Nottingham, University of Nottingham, January 2026. *The potential of hierarchical inference with extreme mass-ratio inspiral observations*.
- Reunion populations de sources GDR ondes gravitationnelles, Institut d'Astrophysique de Paris, September 2025. *Revealing the massive black hole population with LISA*.
- GR24 & Amaldi16, University of Glasgow, UK, July 2025.‡ *Constraints on EMRI Populations from LISA data*.
- EMRI Search & Inference within the LISA Global Fit - Part I, APC Paris, France, June 2025. *Accelerated MBH population estimation from EMRI detections*.
- SUPA Cormack, University of Dundee, UK December 2024. *Constraining Massive Black Hole population from LISA observations*.
- LISA Astrophysics Working Group Meeting, MPA Garching, Germany, Nov 2024. *Constraining EMRI Population using Hierarchical Bayesian Inference*.
- Cosmology from Home, July 2023. *Probing Fuzzy dark matter with lensed Gravitational waves.*
- 3rd International Conference on Condensed Matter and Applied Physics (ICC) - Bikaner, India, October 2019. *Lindblad Evolution and Quantum to Classical Transition of Rabi Oscillation in Single Quantum Dot*.
- 9th International Conference on Gravitation and Cosmology (ICGC) - IISER Mohali, India, December 2019. *Clustering and Predicting Astrophysical events using GW.*

Invited Talks

- Physik-Institut, Universität Zürich, January 2026. *Revealing MBH astrophysics using EMRIs*.
- Institute for Gravitational Research, January 2026. *Revealing the population of massive black holes*.
- LISA Working Group, January 2026. *Constraints on EMRI Populations from LISA data*.
- PyCBC Telecon, April 2025. *Gravitational wave lensing : probing Fuzzy Dark Matter with LISA*.
- LISA Cosmology Working Group, April 2025. *Gravitational wave lensing : probing Fuzzy Dark Matter with LISA*.

Workshops & Summer School organized‡ and attendance

- Python for HPC Workshop by Max Planck Computing and Data Facility (MPCDF), July 2023
- 2^{nd} MaNiTou Summer School on Gravitational Waves : A new window to the Universe - Nice, France, July 2023.

Research Visits

- Visiting researcher, Max-Planck-Institut für Gravitationsphysik (*Albert-Einstein-Institut*), (Duration : January 01 – January 09, 2018 ; Supervisor : Dr M. Zumalacárregui)
- Visiting researcher, Institut Astrophysique de Paris, (Duration : January 01 – January 09, 2018 ; Supervisor : Prof Marta Volonteri)

Awards and Recognition

2025	University of Glasgow Graduate School Mobility Grant
2025	Lord Kelvin or Charles Lindie Mitchell Scholar
2023	University of Glasgow Graduate School scholarship.
2023	Received Erasmus+ funding (EU's program to support education, training, youth and sport in Europe)
2022	Université PSL fellowship for higher education (M2).
2022	Received IPIASMUS grant (IPI Initiative « Physique des Infinis ») towards carrying out an internship at Max-Planck-Institut für Gravitationsphysik (Albert-Einstein-Institut).
2020	Received honorarium by the Max-Planck-Institut für Gravitationsphysik (Albert-Einstein-Institut) for a three-month internship.

Programming & Tools

Python ●●●●●
C++/C ●●●●○
Mathematica ●●●○○
SQL ●●○○○

Scientific Computing

ML / DL ●●●○○
HPC ●●●○○

Cloud Technologies

- **AWS :** S3, CDN, λ

Front-End & Back-End

React + JS ●●○○○
HTML, CSS ●●●○○

Public Libraries

MBHpopulation 2024-Present
github.com/SSingh087/MBHpopulation
Unveiling the population of massive black holes
gravitational-waves | electromagnetic waves | massive black hole | LISA | ZTF | population inference

Poplar 2024-2025
github.com/cchapmanbird/poplar
Augmenting selection bias modelling with machine learning for big speedups
gravitational-waves | massive black hole | LISA | population inference | deep learning

GLoW 2023
github.com/miguelzuma/GLoW_public
Contributed to the GW lensing library
gravitational-waves | lensed-gravitational-waves

Inclusion of higher modes for Parameter estimation of Gravitational waves in PyCBC 2022
github.com/SSingh087/pycbc/tree/conmodel
A recovery model that allows extracting mode-by-mode information while performing parameter estimation.
gravitational-waves | higher-modes | premerger-detection | multi-messenger astronomy | PyCBC

LensGW for generating gravitationally lensed signals 2021
github.com/SSingh087/lensGW
Python library for generating lensed gravitational waves and uses PyCBC for waveform generation so that all analysis can be done using tools provided by PyCBC.
gravitational-waves | lensed-gravitational-waves | PyCBC

lensGW-PyCBC-plugin 2021
github.com/SSingh087/lensGW-PyCBC-plugin
Plugin for allowing waveforms to be recognised by PyCBC and perform parameter estimation
gravitational-waves | lensed-gravitational-waves | PyCBC

LISA - Module 2020
github.com/gwastro/pycbc/commits/master/pycbc/detector.py?author=SSingh087
Prototype for analysis of MBH GWs signals using LISA space-based GW observatory.
The module consists of a simplified LISA orbit and detector response towards a GW signal.
gravitational-waves | LISA | PyCBC

Teaching‡ and Mentorship*,°

- 2024, Physics Lab, P2.

Undergraduate° and master's* students (via Bose.X)

- 2024, Nancy Sharma° & Sree Suswara*, SVNIT, Surat, *Gravitational-Wave Lensing + Machine Learning*.
- 2023, Nancy Sharma° Nancy Jikarda°, SVNIT, Surat, *Gravitational-Wave Lensing*.
- 2021, Adarsh Mahor° & Janvita Reddy°, SVNIT, Surat, *Co-authored a paper; Application of ML in astronomy*.
- 2025, Ankita Chaudhary*, Osmania University,
- 2025, Ansal Kanu°, University of Delhi,

- 2025, Ananthu K Lali*, University of Alabama, Huntsville,
- 2025, Shravan*

Extracurricular & Outreach activities

BOSE.X, Co-Founder :	Independent research organization for undergraduate research ; since 2019. (bosex.org).
GRA-WITTY, Founder :	Outreach program ; since 2023. (gra-witty)
Glasgow Science Festival :	Outreach activity, 2024
Worldcon :	Outreach activity, 2024